CSC1024 PROGRAMMING PRINCIPLES

PYTHON BASED CLASSROOM MANAGEMENT SYSTEM

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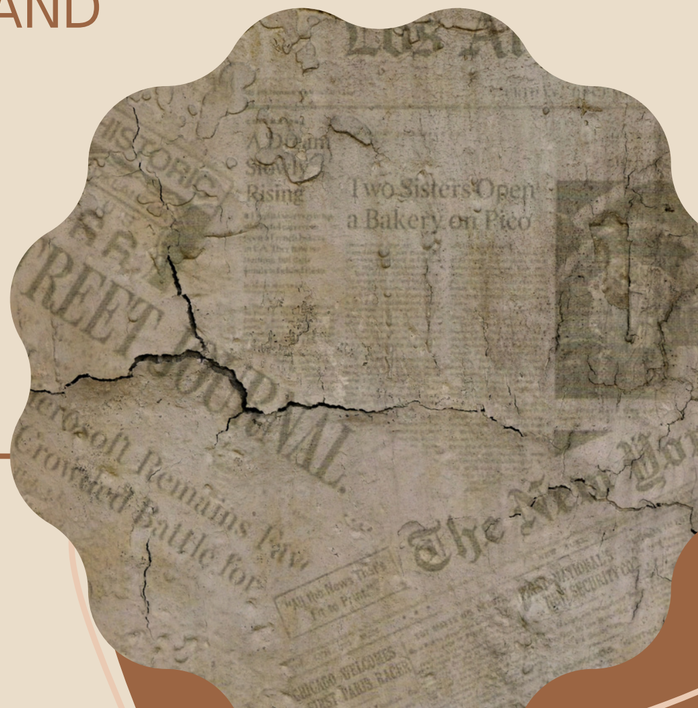
PRESENTATION VIDEO WEB LINK:

https://youtu.be/2HbhivlM5m8?si=ApHgKOu-UDcRu7Wq

OBJECTIVES

- TO DESIGN A USER-FRIENDLY INTERFACE THAT MEETS THE REQUIREMENTS OF TEACHERS STUDENTS AND ADMINISTRATORS
- TO IMPLEMENT FUNCTIONALITIES THAT MANAGES STUDENT RECORDS, ATTENDANCE DATA, GRADES AND SO ON



PROGRAMMING TECHNIQUES

IF-ELSE

```python
def student_login(self):
    student_data = self.read_student_data()
    if not student_data:
        print("No valid student IDs found in the file. Please contact School IT Department @ eduhub@imail.eduhub.edu.my")
        return False

    while True:
        user_input = input("Enter your student ID (type 'quit' to exit): ")
        if user_input.lower() == 'quit':
            print("Exiting...")
            return False
        if user_input in student_data:
            print(f"Name: {student_data[user_input]}, Student ID: {user_input}")
            return True
        else:
            print("Invalid student ID. Please try again.")
```

- if -else statements are used to control the flow of the program
- if the entered student ID is not found in the ‘student data’ dictionary, the else block is executed , which prints “invalid student ID. Please try again.”

IF-ELIF-ELSE

```python
def manage_attendance(self):
    while True:
        print("[1] Teacher Login and Generate Code")
        print("[2] Student Mark Attendance")
        print("[3] Check Attendance")
        print("[4] Exit to main menu")
        choice = input("Enter your choice (1, 2, 3 or 4): ")

        if choice == '1':
            self.teacher_login()
        elif choice == '2':
            self.mark_attendance()
        elif choice == '3':
            self.check_attendance()
        elif choice == '4':
            print("Exiting...")
            break
        else:
            print("Invalid choice. Please enter 1, 2, 3, or 4.")
```

- if function to check whether the selection is equal to 1
- elif function is used when there is additional selection (2,3,4)
- else function is used when the user inputs any other value that is not 1,2,3 or 4, notify the user that the input is invalid

WHILE LOOP

```python
def student_login(self):
    student_data = self.read_student_data()
    if not student_data:
        print("No valid student IDs found in the file. Please contact School IT Department @ eduhub@imail.eduhub.edu.my")
        return False

    while True:
        user_input = input("Enter your student ID (type 'quit' to exit): ")
        if user_input.lower() == 'quit':
            print("Exiting...")
            return False
        if user_input in student_data:
            print(f"Name: {student_data[user_input]}, Student ID: {user_input}")
            return True
        else:
            print("Invalid student ID. Please try again.")
```

- While loop is used to repeatedly prompt the user to enter their student ID until either a valid ID is entered or the user chooses to exit
- While True creates an infinite loop that will continue until a break statement is encountered

TRY-EXCEPT

```python
def calculate_attendance(self, student_id, subject_code):
    total_classes = 25
    attended_classes = 0

    try:
        with open(self.attendance_file, 'r') as f:
            for line in f:
                data = line.strip().split(',')
                if len(data) >= 3:
                    if data[1].strip() == student_id and data[2].strip() == subject_code:
                        attended_classes += 1
    except FileNotFoundError:
        print("Attendance file not found.")
        return None

    attendance_percentage = (attended_classes / total_classes) * 100
    return attendance_percentage
```

- The “try” statement starts a block of code that may potentially raise an exception
- the “try” block attempts to open and read the attendance file
- the “except” block will be executed if the specified file does not exist

TROUBLESHOOTING

PROBLEM 1

HOW TO EXTRACT TWO PARTICULAR DATA FROM A TEXT FILE.

```python
def read_student_data(self):
    student_data = {}
    try:
        with open(self.attendance_file, 'r') as f:
            for line in f:
                student_info = line.strip().split(',')
                if len(student_info) >= 2:
                    name = student_info[0].strip()
                    student_id = student_info[1].strip()
                    student_data[student_id] = name
    except FileNotFoundError:
        print("File not found or path is incorrect.")
    except Exception as e:
        print("An error occurred:", e)
    return student_data
```

- opens the attendance file for reading
- iterate over each line in the file
- split the line by comma to extract student's information
- check if there are at least two elements which is name and ID
- extract and strip the student's name and student's ID
- add student's ID as the key and name as the value to the dictionary

Solution:

- search for information through Google website
- search for the functions that can be used, such as len() from Zeal app

PROBLEM 2

HOW TO RETRIEVE THE CURRENT DATE AND TIME AND FORMATS IT TO RETURN THE ‘WEEKDAY’ INFORMATION

```
if student_code == generated_code:
    student_id = input("Enter student's ID: ")
    student_data = self.read_student_data()
    if student_id in student_data:
        student_name = input("Enter student's full name: ").upper()
        weekday = datetime.now().strftime('%A')

        with open(self.attendance_file, 'a') as f:
            f.write(f"{student_name}, {student_id}, {subject_code}, {datetime.now().strftime('%Y-%m-%d')}, {weekday}\n")
```

- ‘datetime.now( )’ returns a ‘datetime’ object which represents the current date and time
- .strftime(‘%A’): when called on a ‘datetime’ object, it formats the date and time according to the specified format string. ‘%A’ is a directive that represents the full weekday name

Solution:

- seek for teammates advice
- search for a function that can be used to save the ‘weekday’ information

CONCLUSION

- provides an opportunity for students to apply their programming skills in a practical context
- cultivates problem solving abilities and collaboration skills to complete the assignment